

Axiom at DialectSentEval 2026: Dialect-Aware Ensembling of MARBERTv2 and AraBERTv2 for Arabic Sentiment Analysis

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Abstract

Sentiment analysis of regional Arabic dialects presents a formidable challenge due to severe orthographic variation, morphological complexity, and a scarcity of annotated resources. To tackle these linguistic hurdles within Subtask-1 of the DialectSentEval 2026 Shared Task, team **Axiom** proposes a heavily regularized, dialect-aware ensemble framework combining two domain-complementary language models: MARBERTv2 and AraBERTv2. Our pipeline incorporates preprocessed dialect tags appended directly to input sequences, Stochastic Weight Averaging (SWA) for flat-minima convergence, and simple probability averaging across model outputs. Empirical results on the hidden test set demonstrate the robustness of this approach, with our system securing **2nd place** overall and achieving an official **Macro F1-Score of 0.8211**.

1 Introduction

Arabic sentiment analysis has witnessed significant progress with the development of transformer-based Pretrained Language Models (PLMs). However, transferring these models to regional Arabic dialects remains fundamentally challenging. Unlike Modern Standard Arabic (MSA), dialects exhibit diglossia, severe orthographic ambiguity, rich but non-standardized morphology, and frequent code-switching (Farghaly and Shaalan, 2009; Hamed et al., 2025; Alayba, 2025). Because dialectal Arabic is primarily spoken and lacks standardized orthography, the resulting token sparsity and surface-form variations severely degrade the performance of models trained exclusively on formal MSA (Alhafni et al., 2024; Asadi and Asli-Badarneh, 2026).

The DialectSentEval 2026 shared task provides a critical benchmark for evaluating sentiment analysis systems across diverse Arabic dialects (Ezzini et al., 2026a,b). Building upon foundational dialectal datasets, benchmarks, and shared task initiatives

(Alharbi et al., 2025c,a,b), Subtask-1 focuses on 3-class sentiment classification (Negative, Neutral, Positive) over dialectal text sequences.

In this work, we present team **Axiom**'s submission for Subtask-1. We take the stance that when modeling noisy, moderately sized dialectal datasets, optimization stability and domain complementarity are vastly more effective than scaling model capacity or relying on unstable meta-learning stacks. To this end, we propose a highly regularized, zero-added-parameter dual-model ensemble combining MARBERTv2 (Abdul-Mageed et al., 2021) and AraBERTv2 (Antoun et al., 2020). By coupling explicit dialect token prefixing with Stochastic Weight Averaging (SWA) (Izmailov et al., 2018; Lu et al., 2022) to find flatter local optima, our pipeline effectively captures both dialectal slang and formal sentiment cues while heavily resisting overfitting. On the official leaderboard, our system achieved **2nd place** overall with an official **Macro F1-Score of 0.8211** and an **Accuracy of 0.8438**.

Our main contributions are summarized as follows:

- The implementation of a simple yet highly effective mechanism that explicitly conditions transformer attention heads on regional meta-data to mitigate cross-dialect ambiguity.
- The demonstration that integrating Stochastic Weight Averaging (SWA) into the fine-tuning of Arabic PLMs successfully traverses sharp local minima, stabilizing convergence on highly variable text.
- The construction of a robust probability ensemble that seamlessly fuses the informal semantic representations of MARBERTv2 with the structural MSA bias of AraBERTv2, delivering superior generalization without the overhead of stacked architectures.

The remainder of this paper is organized as follows: Section 2 reviews the linguistic challenges and modeling strategies in Arabic dialect NLP. Section 3 outlines our preprocessing, base architectures, and optimization framework. Section 4 presents the experimental setup, official leaderboard results, and ablation analysis. Finally, Section 5 concludes the paper.

2 Related Work

2.1 Linguistic and Orthographic Challenges

Dialectal Arabic sentiment classification is computationally more challenging than MSA processing due to structural diglossia and non-standardized spelling conventions (Boulesnam and Boucetti, 2025; Jaleel, 2025). Dialects differ from MSA across lexical, morphological, and syntactic dimensions (Inoue et al., 2021b; Joubran-Awadie and Shalhoub-Awwad, 2023). Because social media users write dialects phonetically without fixed orthographic standards, token-based NLP models suffer from severe data sparsity (Moaiad et al., 2024; Alnajjar and Hämäläinen, 2024). Consequently, sentiment-bearing adjectives, idioms, and negation patterns frequently appear in forms unobserved by MSA-trained models (Alshemali, 2026; Khalak et al., 2026), underscoring the necessity of dialect-aware normalization and modeling frameworks (Iwidat and Helou, 2026).

2.2 Base Model Selection and Cross-Dialect Transfer

Empirical evidence in Arabic PLM transfer demonstrates that variant proximity between pretraining data and downstream task variety often outweighs raw model size (Inoue et al., 2021a). For informal social media text, MARBERT pretrained on a massive Arabic Twitter corpus regularly outperforms general MSA models on dialectal sentiment benchmarks (Abdul-Mageed et al., 2021; Nwesri et al., 2025). Conversely, MSA-pretrained encoders such as AraBERT retain superior syntactic and structural representations that generalize remarkably well to unseen dialects and formal expressions (Antoun et al., 2020; Boudad et al., 2023; Muller et al., 2020). Combining these distinct model biases provides a balanced coverage of both informal dialectal expressions and structured sentiment syntax.

2.3 SWA and Ensembling in Low-Resource NLP

Fine-tuning transformer models on small datasets risks converging into sharp, non-generalizable local minima. Stochastic Weight Averaging (SWA) addresses this by averaging SGD trajectory weights during training, effectively guiding the parameters toward wider, flatter regions of the loss landscape that demonstrate superior out-of-distribution generalization (Izmailov et al., 2018; Lu et al., 2022; Demir et al., 2024). Furthermore, ensembling diverse PLMs provides proven gains in uncertainty estimation and predictive robustness (Arango et al., 2024). Prior works in dialectal sentiment analysis confirm that multi-model transformer ensembles consistently outperform single-checkpoint baselines (Mansour et al., 2025; Boughareb et al., 2025), validating our choice of probability-averaged ensembling under tight dataset constraints.

3 Methodology

To address the challenges of dialectal variation and the risk of overfitting on a moderately sized dataset, we eschewed complex meta-learning architectures in favor of a highly regularized, dialect-aware ensemble. Our pipeline consists of four primary stages: text preprocessing, base model selection, fine-tuning via Stochastic Weight Averaging (SWA), and probability-based ensembling.

The high-level data flow of this pipeline is illustrated in Figure 1. As depicted, raw dialectal input is first normalized and augmented with regional tags before branching into two parallel, domain-specific transformer encoders, ultimately fusing their regularized probabilistic outputs to produce the final sentiment classification.

3.1 Preprocessing and Representation

Standard Modern Standard Arabic (MSA) preprocessing often aggressively filters characters, which can inadvertently destroy crucial dialectal markers. We utilized the ArabertPreprocessor to clean the text while preserving essential morpho-syntactic boundaries.

To explicitly condition the transformer architectures on regional dialects, we appended a specific dialect token as a prefix to the input sequence. The final input representation is formulated as:

$$x_{\text{input}} = \text{'[' || dialect || '] ' || preprocess}(s) \quad (1)$$

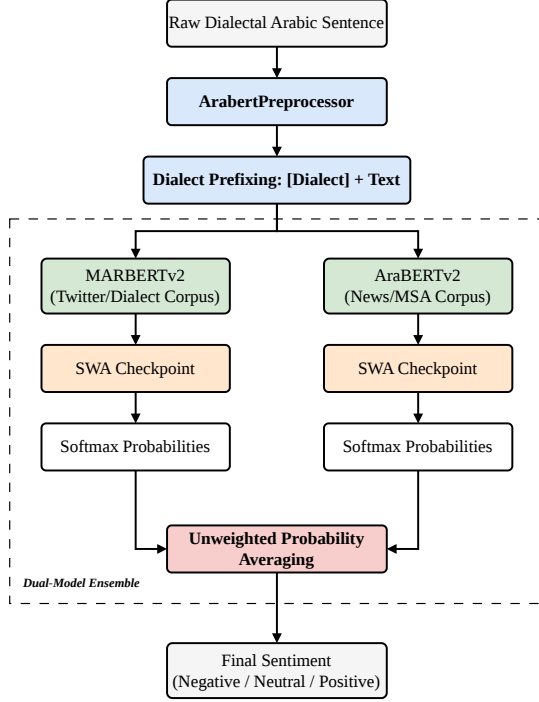


Figure 1: Overall architecture of the Axiom dialect-aware sentiment analysis pipeline.

where s denotes the raw input sentence and $\parallel$ denotes string concatenation. This prefixing mechanism acts as a soft prompt, guiding the attention heads to contextualize subsequent vocabulary based on the indicated regional dialect.

3.2 Base Architectures

We selected two robust, domain-complementary Transformer encoders to ensure representational diversity within our ensemble:

- **MARBERTv2** (Abdul-Mageed et al., 2021): Pretrained heavily on a massive corpus of Arabic tweets, providing strong zero-shot comprehension of informal phrasing, missing diacritics, and regional slang.
- **AraBERTv2** (Antoun et al., 2020): Pretrained predominantly on news corpora, offering a rigid understanding of structural MSA and formal sentiment expressions that often anchor longer sequences.

3.3 Fine-Tuning and Optimization

To prevent data leakage, we partitioned the training data into a 90/10 stratified split based on the 3-class sentiment labels. Both models were fine-tuned independently using standard Cross-Entropy

loss, as margin-based losses (e.g., Focal Loss) were empirically found to overfit on noisy labels in the training set.

To smooth the loss landscape and prevent convergence into sharp local minima, we applied Stochastic Weight Averaging (SWA) (Izmailov et al., 2018). Complete hyperparameter configurations, including learning rate schedules, SWA transition dynamics, and batch statistics, are detailed in Appendix A.

3.4 Ensembling Strategy

For the final predictive framework, we relied on unweighted probability averaging rather than a stacked meta-learner. The output logits from the SWA checkpoints of both models were converted into probability distributions via a Softmax layer. The final sentiment probability was calculated as:

$$P_{\text{ens}}(y|x) = \frac{1}{2} [P_{\text{MAR}}(y|x) + P_{\text{Ara}}(y|x)] \quad (2)$$

The final predicted class $\hat{y}$ corresponds to $\text{argmax}_y P_{\text{ens}}(y|x)$.

4 Experiments and Results

4.1 Experimental Setup

All models were trained using PyTorch with Automatic Mixed Precision (AMP) on an NVIDIA GPU environment. To facilitate exact replication, comprehensive hardware details, sequence lengths, and optimization configurations are provided in Appendix A.

4.2 Local Validation

During local evaluation on our 10% stratified hold-out set ($N = 174$), both models demonstrated strong convergence. MARBERTv2 achieved a standalone Macro F1 of 0.9410, while AraBERTv2 achieved 0.8962. By ensembling both models via probability averaging, the local Macro F1 score peaked at **0.9560**, demonstrating the complementary nature of the two architectures.

4.3 Official Shared Task Results

Our proposed ensemble was evaluated blindly on the official CodaLab test set for Subtask-1. The **Axiom** system achieved **2nd place** overall, recording an official **Macro F1-Score of 0.8211**, an Accuracy of 0.8438, Precision of 0.8188, and Recall of 0.8242.

For context within the competition overview, the first-place system achieved a Macro F1 of 0.8667,

while our ensemble comfortably outperformed the third-place baseline (Macro F1: 0.8175) and the rest of the participating field. These results confirm that our dialect-aware preprocessing paired with flat-minima optimization successfully generalized to unseen dialectal texts, securing a top-tier rank without relying on external datasets or complex meta-learners.

4.4 Ablation and Analysis

To quantify the contribution of our architectural decisions, we tracked the performance of individual components on the official CodaLab blind test set. Table 1 illustrates how our ensembling strategy compares to standalone models and alternative stacking methods. A comprehensive breakdown of all secondary metrics (Accuracy, Precision, Recall) is available in Appendix B.

Model Configuration	Macro F1
MARBERTv2 (Standalone)	0.8096
AraBERTv2 (Standalone)	0.7646
CAMeLBERT-DA (Standalone)	0.7656
Dual-Model Ensemble (Ours)	0.8211
Ensemble + LR Meta-Learner	0.8134
3-Model Ensemble (+ CAMeLBERT)	0.7932

Table 1: Summary of official test scores. Detailed metrics are provided in Appendix B.

Base Model Performance: In isolation, MARBERTv2 proved to be the strongest standalone architecture (0.8096 F1), validating the hypothesis that pretraining on social media text aligns better with the noisy dialectal target domain. Conversely, AraBERTv2 and CAMeLBERT-DA struggled to generalize when acting alone.

The Power of Complementary Ensembling: Despite AraBERTv2’s weaker standalone performance, fusing its probabilistic outputs with MARBERTv2 yielded our highest official score, culminating in an F1-Score of **0.8211**. This confirms that AraBERT captures distinct structural or formal sentiment cues that MARBERT misses, making them highly complementary.

Detrimental Additions: Throughout development, we observed that adding complexity often harmed generalization. Replacing our simple probability average with a Logistic Regression stacking meta-learner degraded the F1 score to 0.8134, likely due to the meta-learner overfitting the highly confident out-of-fold predictions from the small training set. Similarly, introducing CAMeLBERT-DA as a third model introduced destructive noise.

This indicates that misaligned pretraining vocabularies can actively harm small-scale probability ensembles, reinforcing the necessity of strict model curation.

4.5 Error Analysis

While our ensemble generalized exceptionally well locally, a qualitative review of the misclassified validation samples revealed three primary failure modes driven by linguistic complexity. **(1) Subjective Neutrality:** Texts containing implicit complaints alongside neutral anchors were often aggressively predicted as *Negative* by both models, despite being ground-truthed as *Neutral*. **(2) Conflicting Polarities:** Sentences balancing negative aspects with positive justifications confused the models, causing the ensemble to default to *Neutral*. **(3) Negated Negativity:** Formal phrases implying a positive experience through the negation of a negative were conservatively misclassified as *Neutral*. A detailed breakdown with specific Arabic text examples is provided in Appendix C.

5 Conclusion

In this paper, we described team Axiom’s 2nd-place submission for Subtask-1 of the DialectSentEval 2026 Shared Task. We demonstrated that for highly dialectal Arabic sentiment analysis on constrained datasets, a heavily regularized dual-model ensemble consisting of MARBERTv2 and AraBERTv2 yields highly competitive results. By combining explicit dialect tagging with Stochastic Weight Averaging and unweighted probability averaging, our system effectively mitigated overfitting while maximizing representational diversity between social media and formal MSA distributions. Future work will explore the integration of generative data augmentation or adversarial training methods to further close the performance gap on rare dialectal structures without degrading model calibration.

6 Limitations

Despite its strong empirical performance, our approach presents three primary limitations. First, ensembling MARBERTv2 and AraBERTv2 inherently doubles inference latency and memory overhead compared to standalone models. Second, our maximum sequence length of 128 tokens risks truncating long-range context in extended dialectal narratives. Finally, our dialect prefixing mechanism relies on gold-standard regional metadata; deploy-

ing this pipeline in the wild would necessitate an upstream dialect identification module, potentially introducing cascading classification errors.

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References

- Muhammad Abdul-Mageed, AbdelRahim Elmadany, and El Moatez Billah Nagoudi. 2021. [Arbert & marbert: Deep bidirectional transformers for arabic](#). *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 7088–7105.
- Abdulaziz M. Alayba. 2025. [Arabic natural language processing \(nlp\): A comprehensive review of challenges, techniques, and emerging trends](#). *Computers*, 14:497.
- Bashar Alhafni, Sarah Al-Towaity, Ziyad Fawzy, F. Nassar, Fadhl Eryani, Houda Bouamor, and Nizar Habash. 2024. [Exploiting dialect identification in automatic dialectal text normalization](#). *Proceedings of the ArabicNLP Conference*, pages 42–54.
- Maram Alharbi, Saad Ezzini, Hansi Hettiarachchi, Tharindu Ranasinghe, and Ruslan Mitkov. 2025a. Evaluating large language models on arabic dialect sentiment analysis. In *Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing (RANLP)*.
- Maram I Alharbi, Salmane Chafik, Saad Ezzini, Ruslan Mitkov, Tharindu Ranasinghe, and Hansi Hettiarachchi. 2025b. Ahasis: Shared task on sentiment analysis for arabic dialects. In *Proceedings of the Shared Task on Sentiment Analysis for Arabic Dialects*, pages 1–6.
- Maram I Alharbi, Saad Ezzini, Hansi Hettiarachchi, Tharindu Ranasinghe, and Ruslan Mitkov. 2025c. Ador: Dataset for arabic dialects in hotel reviews: A human benchmark for sentiment analysis. In *Proceedings of the First Workshop on Advancing NLP for Low-Resource Languages*, pages 187–191.
- Khalid Alnajjar and Mika Härmäläinen. 2024. [Normalization of arabic dialects into modern standard arabic using bert and gpt-2](#). *Journal of Data Mining and Digital Humanities*, 2024.
- Basemah Alshemali. 2026. [Dialectal substitution as an adversarial approach for evaluating arabic nlp robustness](#). *Scientific Reports*, 16:5996.
- Wissam Antoun, Fady Baly, and Hazem Hajj. 2020. [AraBERT: Transformer-based model for Arabic language understanding](#). In *Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools, with a Shared Task on Offensive Language Detection*, pages 9–15, Marseille, France. European Language Resources Association.
- Sebastian Pineda Arango, Maciej Janowski, Lennart Purrucker, Arber Zela, Frank Hutter, and Josif Grabocka. 2024. [Ensembling finetuned language models for text classification](#). *ArXiv*, abs/2410.19889.
- Ibrahim A. Asadi and Abeer Asli-Badarneh. 2026. [Diglossia and orthographic complexity as multiplicative but not additive challenges in arabic: A critical review](#). *Journal of Psycholinguistic Research*, 55(3):37.
- Naaïma Boudad, R. Faizi, and R. Thami. 2023. [Multilingual, monolingual and mono-dialectal transfer learning for moroccan arabic sentiment classification](#). *Social Network Analysis and Mining*, 14(1):3.
- Djalila Boughareb, Rima Boughareb, Samir Hallaci, Mohammed Raid Elislam Boukherouba, and Hamid Seridi. 2025. [Addressing the complexity of dialectal arabic: An enhanced encoder-decoder ensemble approach for optimized sentiment analysis](#). *ACM Transactions on Asian and Low-Resource Language Information Processing*, 24:1–21.
- Ilhem Boulesnam and Rabah Boucetti. 2025. [Arabic language characteristics that make its automatic processing challenging](#). *International Arab Journal of Information Technology*, 22:814–831.
- Caglar Demir, Arnab Sharma, and A. Ngomo. 2024. [Adaptive stochastic weight averaging](#). *ArXiv*, abs/2406.19092.
- Saad Ezzini, Shadi Abudalfa, Maram Alharbi, Salmane Chafik, Hind Alatawi, Mo El-Haj, Ahmed Abdelali, Osamah Alnahari, and Salima Lamsiyah. 2026a. Dialectsenteval 2026: Arabic dialect sentiment analysis and swapping shared task. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary.
- Saad Ezzini, Shadi Abudalfa, Maram Alharbi, Salmane Chafik, Hamzah Luqman, Mo El-Haj, Paul Rayson, Samaher Alghamdi, and Reem Alotaibi. 2026b. Arasenteval: A shared task on sentiment analysis and swapping in arabic. In *Proceedings of the 7th Workshop on Open-Source Arabic Corpora and Processing Tools (OSACT7), co-located with the Language Resources and Evaluation Conference (LREC 2026)*, Palma, Mallorca, Spain.
- A. Farghaly and Khaled Shaalan. 2009. [Arabic natural language processing: Challenges and solutions](#). *ACM Transactions on Asian Language Information Processing*, 8:1–22.
- Injy Hamed, Caroline Sabty, Slim Abdennadher, Ngoc Thang Vu, Thamar Solorio, and Nizar Habash.

2025. [A survey of code-switched Arabic NLP: Progress, challenges, and future directions](#). In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 4561–4585, Abu Dhabi, UAE. Association for Computational Linguistics.
- Go Inoue, Bashar Alhafni, Nurpeiis Baimukan, Houda Bouamor, and Nizar Habash. 2021a. [The interplay of variant, size, and task type in Arabic pre-trained language models](#). In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 92–104, Kyiv, Ukraine (Virtual). Association for Computational Linguistics.
- Go Inoue, Salam Khalifa, and Nizar Habash. 2021b. [Morphosyntactic tagging with pre-trained language models for arabic and its dialects](#). *Findings of the Association for Computational Linguistics (ACL)*.
- Hani Iwida and Mamoun Abu Helou. 2026. [Arabic dialect nlp: A unified taxonomic, methodological, and trend-driven survey](#). *Journal of Information Technology, Cybersecurity, and Artificial Intelligence*.
- Pavel Izmailov, Dmitrii Podoprikin, T. Garipov, D. Vetrov, and A. Wilson. 2018. [Averaging weights leads to wider optima and better generalization](#). *Uncertainty in Artificial Intelligence (UAI)*.
- T. K. Jaleel. 2025. [Advanced technologies in arabic language communication: Challenges and future directions](#). *ETHOS: Jurnal Penelitian dan Pengabdian kepada Masyarakat*.
- Nancy Joubran-Awadie and Yasmin Shalhoub-Awwad. 2023. [Morphological distance between spoken palestinian dialect and standard arabic and its implications for reading acquisition](#). *First Language*, 43:200–230.
- Abdulmuizz Khalak, Abderrahmane Issam, and Gerassimos Spanakis. 2026. [From FusHa to folk: Exploring cross-lingual transfer in Arabic language models](#). In *Proceedings of the 13th Workshop on NLP for Similar Languages, Varieties and Dialects*, pages 196–209, Rabat, Morocco. Association for Computational Linguistics.
- Peng Lu, Ivan Kobzyev, Mehdi Rezagholizadeh, Ahmad Rashid, Ali Ghodsi, and Phillippe Langlais. 2022. [Improving generalization of pre-trained language models via stochastic weight averaging](#). In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 4948–4954, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Omar Mansour, Eman Aboelela, Remon Talaat, and Mahmoud Bustami. 2025. [Transformer-based ensemble model for dialectal arabic sentiment classification](#). *PeerJ Computer Science*, 11.
- Yazeed Al Moaiad, Mohammad Alobed, Mahmoud Al-sakhnini, and Alaa M. Momani. 2024. [Challenges in natural arabic language processing](#). *Edelweiss Applied Science and Technology*.
- Benjamin Muller, Benoît Sagot, and Djamé Seddah. 2020. [Can multilingual language models transfer to an unseen dialect? a case study on north african arabizi](#). *ArXiv*, abs/2005.00318.
- Abdusalam F. Ahmad Nwesri, Nabila Almabrouk S. Shinbir, and Amani Bahlul Sharif. 2025. [Sentiment analysis on Arabic dialects: A multi-dialect benchmark](#). In *Proceedings of the Shared Task on Sentiment Analysis for Arabic Dialects*, pages 86–91, Varna, Bulgaria. INCOMA Ltd., Shoumen, Bulgaria.

A Implementation and Hyperparameter Details

To facilitate reproducibility, we provide the exact hyperparameters utilized during the training phase. All models were trained utilizing a maximum sequence length of 128 tokens and a batch size of 16. We employed the AdamW optimizer with a base learning rate of 2×10^{-5} and a weight decay of 0.01. A linear learning rate scheduler with a 10% warmup period was applied during the initial phase.

Stochastic Weight Averaging (SWA) (Izmailov et al., 2018) was activated at epoch 3, transitioning to an SWA-specific learning rate of 1×10^{-5} . The SWA weights were updated according to the following schedule:

$$w_{\text{SWA}} \leftarrow \frac{w_{\text{SWA}} \cdot n_{\text{models}} + w}{n_{\text{models}} + 1} \quad (3)$$

Batch Normalization statistics were updated across the entire training loader at the conclusion of the 10th epoch. All training was conducted using PyTorch with Automatic Mixed Precision (AMP) to optimize VRAM utilization on an NVIDIA GPU.

B Detailed Ablation Metrics

Table 2 provides the comprehensive evaluation metrics (Accuracy, Precision, Recall, and Macro F1) for all tested configurations on the official blind test set.

C Qualitative Error Analysis Examples

Table 3 provides specific examples from our local 10% validation holdout where the dual-model ensemble failed to predict the correct sentiment. These examples illustrate the primary failure modes discussed in Section 4.5, specifically highlighting the difficulty of resolving conflicting polarities and implicit sentiment in dialectal Arabic.

Model Configuration	Accuracy	Precision	Recall	Macro F1
<i>Standalone Base Models:</i>				
MARBERTv2	0.8324	0.8067	0.8138	0.8096
CAMeLBERT-DA	0.7829	0.7667	0.7765	0.7656
AraBERTv2	0.7886	0.7652	0.7700	0.7646
Dual-Model Ensemble (Ours)	0.8438	0.8188	0.8242	0.8211
<i>Alternative Formulations:</i>				
Dual-Ensemble + LR Meta-Learner	0.8305	0.8121	0.8207	0.8134
3-Model Ensemble (+ CAMeLBERT)	0.8133	0.7920	0.8012	0.7932

Table 2: Comprehensive ablation study of official test scores, demonstrating the impact of base model selection and various ensembling strategies.

Raw Arabic Input & Translation	True	Pred.	Failure Mode
متوسط، ممكن يكون أحسن و توقعنا ان الغرف والحمامات تكون أنظف قضينا يومين من ليايلنا الأولى في القاهرة هناك (Average, could be better and we expected the rooms and bath- rooms to be cleaner...)	Neutral	Negative	Subjective Neutrality / Implicit Complaint
بالرغم من بلي فندق كايغطي شي ثمن كبيرا مقارنة بالحيران، و لكن أنا كنت بيه عشان كان مشهور و كان عندو تقييمات. (Even though the hotel charges a high price compared to neigh- bors, I stayed because it was famous...)	Negative	Neutral	Conflicting Polarities (Mixed Sentiment)
وصلنا هناك قبل يوم من الموعد المحدد، لكن هذا لا مشكلة في فهمي. (We arrived there a day before the scheduled time, but this is no problem...)	Positive	Neutral	Negated Negativity (Implicit Positivity)

Table 3: A representative sample of misclassified instances from the local validation set, highlighting the challenges of implicit phrasing and boundary ambiguity in dialectal Arabic.